

Week 7 Final Submission Report

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Project: Backend Development Project

Phase: Phase 2

Week: 7

1. Objective

Week 7 focused on database schema mapping and storage, secure token authorization, database transaction handling, and a Redis caching layer with database fallback.

2. Database Schema and Mapping

A Week 7 migration was added to extend the existing products table with a user_id field. New products are associated with the authenticated user obtained from the JWT. A user-specific endpoint was implemented to retrieve products belonging to the logged-in user.

Migration file: src/migrations/week7Migration.js

User-specific endpoint: GET /api/products/my-products

3. Authentication and Authorization

The existing JWT authentication flow was extended with product ownership authorization. After JWT verification, the application compares the authenticated user's ID with the product's user_id before allowing protected update and delete operations.

4. Transaction Handling

Product updates were implemented using a SQLite transaction flow: BEGIN TRANSACTION, database update, COMMIT on success, and ROLLBACK when an error occurs.

5. Redis Caching

Redis was deployed using Docker and connected to the Node.js application. The Product API uses the cache key products:all with a 60-second expiry. When cached data is unavailable, the application retrieves products from SQLite and stores the result in Redis.

6. Verification Results

Verification	Result
Database migration	PASS
User-product relationship	PASS
JWT authentication	PASS
Product ownership authorization	PASS
Unauthorized-user test	PASS
Transaction-based update	PASS
Redis connection	PASS
Redis cache miss / database retrieval	PASS
Redis cache hit	PASS
Redis unavailable / SQLite fallback	PASS

7. Key Verification Evidence

A repeated GET /api/products request first returned a database source and then returned a cache source, demonstrating Redis cache use. Redis was also stopped temporarily while the Node.js API remained running; the Product API continued to return data from SQLite, demonstrating fallback behavior.

An authenticated user was prevented from modifying a product owned by another user, returning the expected 403 ownership response.

8. Documentation Deliverables

Week 7 documentation was prepared for database/schema mapping, authentication and authorization testing, and verification results.

Recommended files:

docs/week7-database.md
docs/week7-auth-tests.md
docs/week7-verification.md

9. Conclusion

The Week 7 implementation and verification work was completed successfully. The backend now combines JWT-based authorization, user-product ownership, transaction handling, and Redis caching with SQLite fallback.